

Final PDB + AlphaFold Automation

RCSB-Only PDB Search Logic & Independent AlphaFold Logic

Purpose

The final workflow searches PDB structures directly through RCSB using the user's protein name and normalized variants. UniProt is excluded from PDB discovery and PDB target validation. UniProt is retained only for the independent AlphaFold branch.

Core rule: a protein mention in a PDB title or contextual description is not sufficient. The actual deposited polymer entity must support the requested target.

1. Overall Architecture

```
USER INPUT
|
+---- PDB BRANCH -----+
| |
| user name + normalized variants |
| | |
| v |
| RCSB PDB search |
| | |
| v |
| broad candidate PDB IDs |
| | |
| v |
| deposited polymer entities |
| | |
| human + protein + X-ray |
| | |
| v |
| target-entity validation |
| | |
| v |
| PDB data + scoring |
+-----+

INDEPENDENT ALPHAFOLD BRANCH
user input -> UniProt enrichment -> AlphaFold DB
```

The branches are independent. A UniProt failure must not prevent PDB discovery, and the presence of PDB structures does not prevent AlphaFold retrieval.

2. User Input & Query Variants

The user supplies a protein name, gene symbol, recognized alias, or accession. The PDB branch starts from this supplied text. Normalized variants include the original form, spacing/hyphen variants, and normalized whitespace.

For a general/family query such as "**toll like receptor**", the search remains broad enough to discover relevant Toll-like receptor members instead of silently restricting the PDB search to one UniProt-resolved protein.

3. RCSB PDB Candidate Discovery

Two RCSB polymer-entity search fields are used:

Route	RCSB field	Purpose
1	rccb_polymer_entity.pdbx_description	Broad deposited entity-description search
2	rccb_polymer_entity.rccb_polymer_name_combined	Secondary deposited polymer-name search

Search results are paginated and duplicate PDB IDs from different variants/routes are merged before validation.

```
for name in normalized_query_variants(protein_name):
    search RCSB entity description
    search RCSB polymer name
    merge returned PDB IDs

# No UniProt PDB search
# No UniProt-derived gene PDB search
```

4. Mandatory PDB Filters

Filter	Requirement
Target organism	Homo sapiens / taxonomy 9606
Experimental method	X-RAY DIFFRACTION
Polymer type	Protein
Target confirmation	Actual deposited polymer entity must support the query

5. Strict Target-Entity Validation

The validator inspects the deposited polymer entity itself rather than relying on a PDB title. Evidence includes polymer type, source organism, deposited entity/macromolecular names, gene names/synonyms, sequence, chains, and reference accessions as audit metadata.

UniProt accession is not an acceptance gate. If an entity has a UniProt reference, it is retained only as metadata and may later seed the independent AlphaFold branch.

CDK4 example: a PDB describing a CDK inhibitor is not accepted merely because the title mentions CDK4/6. The deposited target entity must support CDK4.

Toll-like receptor example: a family query can accept an actual deposited human Toll-like receptor entity without first choosing TLR2 through UniProt.

6. Explicitly Excluded

- title-only/context-only matches
- ligand-only matches
- assembly descriptions without target-entity confirmation
- non-human target entities
- non-X-ray structures
- non-protein polymers
- structures that cannot be confirmed as the requested deposited target

7. PDB Data Collected After Validation

Category	Examples
Structure	PDB ID, method, resolution
Composition	Deposited residues, modeled residues, residue difference, chains
Refinement	Depositor R-work and R-free
Validation	DCC R-free, clashscore, Ramachandran, rotamer/sidechain, RSRZ
Thermal/geometry	Average B-factor and validation information
Ligands/activity	Ligand IDs/names and available binding-affinity/activity data

8. Precision & Validation Sources

Numeric values are preserved as text where needed so Excel does not alter displayed precision. Decimal arithmetic avoids binary floating-point rounding. Missing values remain N/A.

The workflow uses official wwPDB validation information as the primary source where available, with the RCSB entry API retained as a fallback for relevant validation metrics.

9. Independent AlphaFold Branch

UniProt is used only to enrich the AlphaFold branch. The script attempts to resolve the user input to a human UniProt entry and then retrieve AlphaFold information. Failure or ambiguity in this branch does not filter PDB structures.

PDB acceptance:
RCSB deposited entity -> human + protein + target confirmation
(NO UniProt acceptance gate)

AlphaFold:
user input -> UniProt enrichment -> AlphaFold DB
(independent branch)

10. Final End-to-End Flow

1. Read user protein name
2. Generate normalized PDB query variants
3. PDB branch
 - > RCSB entity-description search
 - > RCSB polymer-name search
 - > merge candidate PDB IDs
 - > fetch entry metadata
 - > require X-RAY DIFFRACTION
 - > inspect deposited polymer entities
 - > require Homo sapiens + protein
 - > confirm target entity
 - > collect structure/validation/ligand data
4. AlphaFold branch
 - > independent UniProt enrichment
 - > AlphaFold DB retrieval
 - > retain AlphaFold even when crystal structures exist
5. Workbook
 - > PDB_Data
 - > Target_Validation
 - > AlphaFold_Data
 - > supporting/run information
6. PDB quality scoring
 - > H1 / H2 / H3
 - > rank structures for SBDD

11. Final Design Principles

Principle	Meaning
RCSB-first discovery	PDB search starts directly from user input.
No UniProt dependency for PDB	UniProt cannot block, select, or validate a PDB.
Broad discovery + strict validation	High recall first; biological specificity afterward.
Entity-level evidence	Deposited polymer entity is the acceptance basis.
Human X-ray only	Target entity must be human and entry must be X-ray.
Family-aware behavior	General terms can retrieve relevant members.
Independent AlphaFold	AlphaFold is handled separately and is collected even when crystal structures exist.
Auditability	Validation evidence and metadata are retained.

Final rule: UniProt helps the AlphaFold branch; RCSB deposited polymer entities determine whether a PDB structure belongs in the PDB dataset.